Analyzing the Impact of Transpose Layers on CNN-Based Deep Learning

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***Abstract*—Precise and real-time pothole detection plays a vital role in intelligent transportation systems, where timely identification of road damage can improve safety and maintenance efficiency. While YOLO (You Only Look Once) models have been widely adopted for object detection tasks, integrating transpose convolutional layers as a learnable upsampling mechanism within modern YOLO architectures remains insufficiently explored. This study investigates the architectural impact of replacing static upsampling layers with ConvTranspose2d in YOLOv5m and YOLOv8m models, particularly for detecting small, irregular potholes under challenging real-world conditions. These include scenarios with high visual complexity, such as potholes partially obscured by water, surface debris, or shadows. By enabling the network to reconstruct fine spatial details adaptively, ConvTranspose2d proves beneficial for improving localization accuracy in these complex environments. Experimental results demonstrate that adding transpose layers enhances detection performance across both models. YOLOv5m with transpose layers achieved a mAP@50 of 0.839, indicating a 1.57% improvement over the baseline, while YOLOv8m showed a larger relative gain from 0.751 to 0.777 (3.46%). YOLOv5m maintained a more favorable balance between precision (0.894) and inference latency, whereas YOLOv8m demonstrated higher generalization, reflected in an increase in recall from 0.640 to 0.720, albeit with increased inference time. These findings offer valuable insights into architecture-aware enhancements and highlight the potential of ConvTranspose2d for real-time, safety-critical object detection tasks, especially in visually complex road monitoring scenarios.**

Keywords—Pothole Detection, YOLO, Convolutional Neural Network, Transpose Convolutional Layer, Deep Learning.

I. INTRODUCTION

Highways play a crucial role in supporting economic growth and trade. However, they are highly susceptible to surface damage caused by heavy vehicle loads, poor drainage, and extreme weather. Among the most dangerous forms of deterioration is the formation of potholes, which can cause traffic accidents, vehicle damage, and economic losses. Traditional detection methods, such as manual inspections or vehicle-based surveys, are time-consuming and labor-intensive and often fail to detect small or early-stage defects [1]. In response, computer vision technologies, particularly those powered by deep learning, have emerged as promising tools for automating road surface monitoring and damage detection [2].

One of the main focuses in the development of computer vision systems is object detection [3], especially for applications that require real-time object identification and tracking of object detection [4], [5]. Convolutional Neural Networks (CNN) have become the foundation of modern object detection systems, enabling remarkable progress in identifying and localizing visual features [6], [7]. Among CNN-based detectors, the YOLO family has stood out for its speed and accuracy in the real-time detection task [8].

Continued development of YOLO architectures is essential to incorporate more advanced learning strategies, adapt to diverse data, and achieve higher precision. Specifically, enhancing spatial representation and feature learning remains a key challenge in detecting small or subtle objects like potholes [9], [10], [11], [12]. One promising direction is the integration of transpose convolutional layers (ConvTranspose2d) into the detection pipeline. Unlike fixed interpolation methods like nearest neighbor or bilinear, transpose convolutions enable learnable upsampling, allowing the model to reconstruct finer spatial details and better adapt to feature complexity in road imagery [13], [14].

In this study, we explore the effect of inserting transpose layers into two popular YOLO architectures: YOLOv5m and YOLOv8m [15]. We compare the performance of the original models with their modified counterparts that include transpose layers in the head section [16]. Evaluation is conducted across multiple metrics, including detection accuracy (mAP), inference speed, and robustness under diverse visual conditions to reflect realistic and complex road scenarios, including potholes with varying shapes, lighting conditions, occlusions, and surface noise. These characteristics make it ideal for assessing how well learnable upsampling layers like ConvTranspose2d improve detection performance in visually challenging environments. While the focus of this work is pothole detection, the approach is generalizable and can be applied to other domains such as crack detection, medical imaging, or defect identification in industrial settings, or any scenario where high-resolution feature reconstruction is critical [17], [18].

II. LITERATURE REVIEW

Object detection using deep learning has advanced rapidly, with the YOLO architecture recognized for its real-time performance and accuracy. While early versions like YOLOv3 [19] performed well, YOLOv5 and YOLOv8 have emerged as the most robust and efficient models in various

studies [9], [10], [15], [16]. Numerous researchers have focused on optimizing these models further. Liu et al. [9] and Chandra et al. [10] validated YOLOv5's effectiveness in detecting road defects, citing its high inference speed and accuracy. Similarly, Kumar et al. [12] demonstrated the model's viability in real-time road monitoring. Recent research has also highlighted YOLOv8's improvements in architectural design along with processing speed, establishing it as a highly competitive model [15], [20].

To enhance the detection of small, irregular, or partially occluded objects such as potholes, recent studies have explored using learnable transpose convolution layers (ConvTranspose2d) to replace traditional static upsampling techniques. These layers offer improved spatial detail reconstruction and richer feature representations [21], [22]. For instance, Helen et al. applied them in dense convolutional architectures for a classification task [14], while Sestito et al. demonstrated their effectiveness in FPGA-based image reconstruction systems [13]. These prior works confirm the potential of transpose layers in enhancing feature representations, although their use in object detection models like YOLO remains limited. Most existing research tends to focus on improving a single version of YOLO without examining how such architectural modifications perform across different model generations. Given that YOLOv5 and YOLOv8 are widely recognized for their optimal trade-offs between accuracy and real-time performance [11], [20], it is crucial to investigate which architecture gains more from the integration of ConvTranspose2d, especially in the context of detecting subtle and low-contrast road defects under complex visual conditions.

This study addresses the research gap by directly comparing the effects of transpose layers on YOLOv5 and YOLOv8 under identical conditions. Rather than optimizing a single model, it investigates how architectural differences impact the effectiveness of learnable upsampling in pothole detection. As highlighted by Zhang et al. [15] and Shakhovska et al. [23], architectural customization plays a key role in enhancing sensitivity to fine-grained features. By conducting a comparative evaluation, this work contributes novel insights into architecture-aware enhancement strategies for real-time, safety-critical road monitoring systems

III. METHODOLOGY

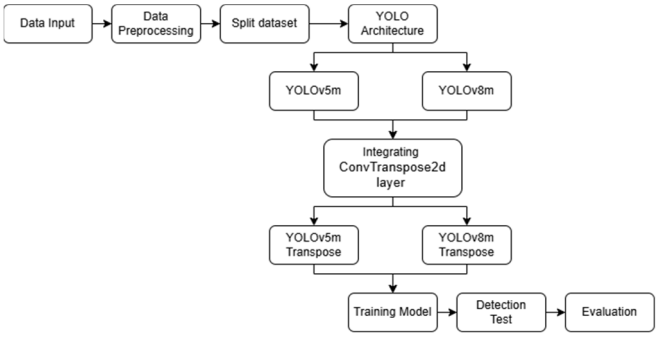


Fig. 1. Research Workflows

As illustrated in Figure 1, this section outlines the comprehensive methodological workflow for evaluating and comparing the performance of both baseline and modified YOLOv5 and YOLOv8 models, which incorporate ConvTranspose2d layers. The process begins with data input, followed by preprocessing steps including image resizing, normalization, and annotation formatting to prepare the dataset for model training. Subsequently, the dataset is systematically divided into training, validation, and testing subsets to ensure robust and consistent evaluation across all model variants.

A. Model Architecture

This study compares the performance of YOLOv5m and YOLOv8m models, with and without transpose convolution layers, for pothole detection. Figures 2 and 3 show that both models follow the standard YOLO architecture: a backbone for feature extraction, a neck for feature fusion, and a head for object prediction. YOLOv5 uses CSPDarknet and PANet, while YOLOv8 adopts a more modern design with BaseBackbone, BaseYOLONeck, and BaseDenseHead for improved multi-scale representation.

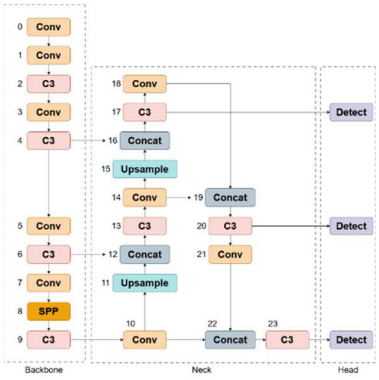


Fig. 2. YOLOv5 Model Design

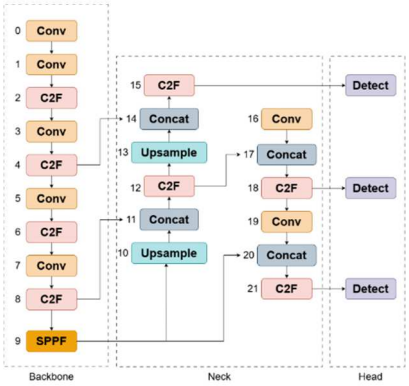


Fig. 3. YOLOv8 Model Design

We replaced the original upsampling layers in both models with ConvTranspose2d to evaluate the effect of learnable upsampling. These layers reconstruct higher-resolution feature maps, improving the model's capability to capture detailed spatial information, which is crucial for detecting small or irregularly shaped potholes.

B. Transpose Layers Integration Strategy

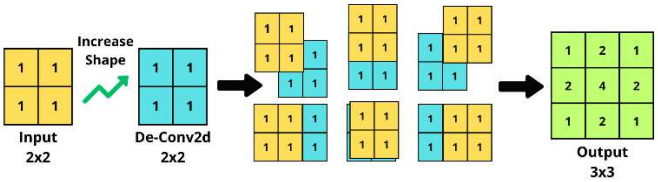


Fig. 4. ConvTranspose2d Layer

YOLO employs CNN architecture, which consists of three main components: the backbone, neck, and head. This study explores the impact of replacing the standard upsampling layers with nn.ConvTranspose2d in two modern object detection models, YOLOv5m and YOLOv8m. This modification is applied in the Head section, where multi-scale feature maps are fused through upsampling operations.

Unlike traditional upsampling methods such as nearest neighbor or bilinear interpolation, which rely on fixed interpolation rules, ConvTranspose2d (Figure 4) performs learnable upsampling by reversing the convolution process. In standard convolution, feature extraction reduces spatial resolution through kernel sliding and aggregation. Conversely, transpose convolution expands the spatial dimensions by inserting padding (usually zeros) between input elements, effectively creating a larger intermediate feature map before applying convolution using trainable filters. This expansion allows the network to learn how to generate higher-resolution feature maps tailored to the specific features in the training data.

Technically, during ConvTranspose2d, each input pixel contributes to multiple output pixels based on the learned weights in the kernel, overlapping contributions from adjacent pixels. These overlapping regions are then aggregated (summed), enabling the network to reconstruct sharp boundaries, fine textures, and detailed object shapes. The process preserves contextual information better than fixed interpolation, especially in complex scenes. This mechanism proves especially valuable in object detection tasks where target features are small, irregular, or partially occluded, such as potholes with water pooling, shadows, or noise artifacts. Because the upsampling is learnable, the model can adapt to the data distribution, resulting in more accurate localization and improved recall of subtle defects [21], [22], [24].

Experimental results confirm that integrating ConvTranspose2d improves localization accuracy and object recall compared to static upsampling. These findings align with earlier studies, including Helen et al. [14], who conceptually discussed the role of transposed convolutions in dense classification settings, and Sestito et al. [13], who demonstrated their effectiveness in preserving spatial detail within FPGA-based image reconstruction systems. While these earlier works highlighted transpose layers' theoretical potential and hardware-focused advantages, they did not assess their impact within real-time object detection frameworks.

To maximize the benefits of this integration, careful design of the ConvTranspose2d configuration is required. Ensuring that the output dimensions align with the skip-connected feature maps is essential for maintaining architectural consistency [25]. The parameter selection strategy in this study includes:

1) Stride and Kernel

The stride is set to 2 to mimic the behavior of the original Upsample layer, which doubles the spatial resolution. A kernel size of 4 is used for P3 and P4 scales, while P5 employs a kernel of size 2 to minimize artifacts and maintain computational efficiency.

2) Padding and Output Padding

Padding is configured as 1, and output_padding is also set to 1. This configuration satisfies the output dimension formula:

$$H_{out} = (H_{in} - 1) \times stride - 2 \times padding + kerneL_size + output_padding, \quad (1)$$

These settings ensure that the upsampled feature maps have dimensions compatible with the corresponding skip-connected layers, avoiding shape mismatch errors during concatenation.

3) Output Channels

As detailed in Table 1, the ConvTranspose2d layers preserve the original channel dimensions at each detection scale (256 for P3 and 512 for P4), ensuring smooth integration with the detection head and maintaining YOLO's multi-scale design.

TABLE I. YOLO V5+ TRANSPOSE HYPERPARAMETER SETUP

Detection Scale	Kernel Size	Stride	Padding	Output Padding	Output Channel
P3	2	2	0	-	256
P4	2	2	0	-	512

TABLE II. YOLOV8 + TRANSPOSE HYPERPARAMETER SETUP

Detection Scale	Kernel Size	Stride	Padding	Output Padding	Output Channel
P3	3	2	1	1	192
P4	3	2	1	1	384

These parameter settings ensure that the spatial dimensions of the upsampled feature maps align with their corresponding high-resolution skip connections, avoiding shape mismatch during concatenation and preserving the integrity of feature fusion.

C. Dataset

The dataset of the road pothole images is from Kaggle [26]. The dataset contains a total of 2105 images. The dataset is one in which labels are given to each image obtained. After downloading and extracting these datasets, it has to be prepared in a proper dataset manner, so that during evaluation and training, the model learns properly.

D. Preprocessing Data

The process involves the steps as presented in the following:

1) Dataset Splitting

The dataset will be partitioned into a validation set containing 16 images and a training set containing 2067 images after preprocessing. This partitioning assists in learning so that the model can be properly trained and evaluated.

2) Auto-orientation of pixel data

Images taken from various devices may have EXIF metadata that stores the image orientation. To ensure that all images have consistent orientation, automatic orientation is applied, and the EXIF orientation information is removed, rotating the images to the correct orientation.

3) Resize to 640x640 (Stretch)

To ensure uniformity of all images used during model training, the dimensions of each image are adjusted to 640x640 pixels. If an image does not match the desired size, it is stretched in one dimension.

4) Augmentation

The following image augmentations are applied to produce three different versions of each image: horizontal and vertical flips with a 50% probability, 90° rotations in both clockwise and counterclockwise directions, random cropping within a range of 0% to 20%, random rotations from -15° to +15°, random horizontal shearing, brightness and exposure modifications within ±25%, application of Gaussian blur between 0 and 5.25 pixels, and addition of salt-and-pepper noise affecting 5% of the pixels.

E. *Training and Evaluation Metrics*

All models in this study were trained under identical conditions and evaluated using standardized metrics commonly applied in YOLO-related research, including Precision, Recall, mAP@50, and mAP@50-95. These metrics follow the evaluation protocols established in widely used datasets such as COCO and Pascal VOC, allowing consistent comparison with existing literature. All experiments were conducted using identical hardware environments to ensure reproducibility and minimize performance variance [27]. The performance of each variant was assessed using several key evaluation metrics:

1) *Precision*

This metric determines how many true positive detections the model achieves relative to all positive predictions, reflecting its ability to detect potholes accurately.

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

2) *Recall*

This metric evaluates the percentage of real potholes the model successfully detects, showing how effectively it captures all relevant cases.

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

3) *mAP50 (Mean Average Precision)*

This metric calculates the average precision across all object categories and detections, considering only those where the intersection between predicted and actual bounding boxes meets or exceeds 50%.

4) *mAP50-95*

This metric offers a more thorough and strict assessment by averaging the precision scores over IoU thresholds from 0.50 to 0.95 in steps of 0.05, resulting in a more detailed evaluation of the model's accuracy.

5) *Inference Time (ms)*

This metric refers to the duration it takes for the model to generate predictions from a single image. This includes the forward pass through the neural network, excluding preprocessing and postprocessing stages.

6) *Post-Processing Time (Non-Maximum Suppression)*

This metric refers to the duration required to filter overlapping bounding boxes after predictions. YOLO generates overlapping boxes for the same object, and NMS ensures that only the most confident prediction is retained.

IV. RESULT AND DISCUSSION

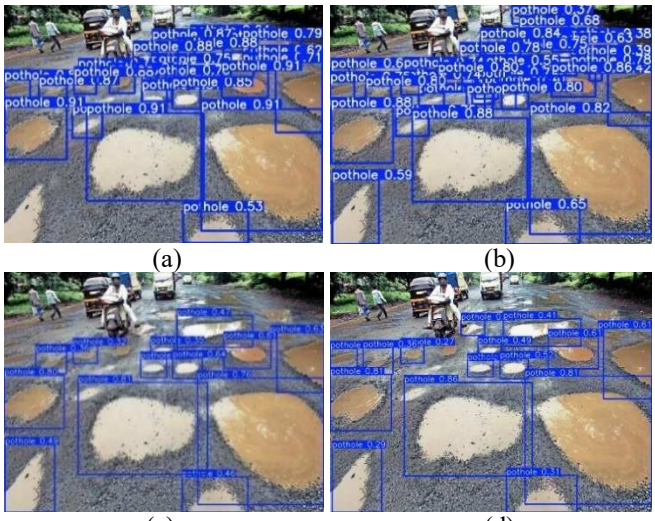
This section evaluates four YOLO variants: YOLOv5m, YOLOv5m with ConvTranspose2d, YOLOv8m, and YOLOv8m with ConvTranspose2d across static image detection, video analysis, and performance metrics. The objective is to examine the effect of replacing static upsampling with learnable transpose convolution on detecting small and irregular potholes. The models were tested on a visually complex dataset containing occlusion, water accumulation, and background noise to simulate real-world conditions and evaluate their ability to reconstruct fine spatial details accurately.

A. *Static Image Detection Results*

The comparative detection results are summarized below. Fig. 5 provides an example of the input image, illustrating the raw imagery used to evaluate model performance.



Fig. 5. Input Image



(a) (b) (c) (d)

Fig. 6. (a)Default YOLOv5m Image Detection, (b)YOLOv5m Image Detection with Transpose Layer, (c)YOLOv8 Image Detection, (d)YOLOv8 Image Detection with Transpose Layer

The results highlight distinct detection behaviors across architectures. In Figure 6(a), the baseline YOLOv5m detected 25 potholes with an inference time of 124.6 ms, aided by minimal preprocessing (0.6 ms) but a relatively high NMS time (131.6 ms). Upon integrating ConvTranspose2d layers in Figure 6(b), YOLOv5m detected 36 potholes, an increase of 44%, indicating superior ability to detect small and fragmented potholes. Inference time slightly decreased to 122.2 ms, although NMS increased to 150.0 ms due to the higher number of predictions. Meanwhile, YOLOv8m (Figure 6c) achieved the fastest inference (112.7 ms) and minimal post-processing time (1.6 ms), yet detected only 13 potholes, revealing limitations in capturing low-contrast or irregular features. The addition of ConvTranspose2d (Figure 6d) improved detection to 15 potholes, confirming the benefit of learnable upsampling, albeit with increased inference time (152.2 ms). These results suggest that ConvTranspose2d

enhances fine-grained feature recognition, especially in YOLOv5m, while in YOLOv8m, the improvement is more moderate but still valuable for small object detection tasks.

B. *Video-Based Detection Results*

This study also extends pothole detection to real-time video streams, showing robust performance across dynamic frames with varying lighting and environmental conditions. Snapshot images were captured from the video, as shown in Figure 7.



Fig. 7. Raw Image

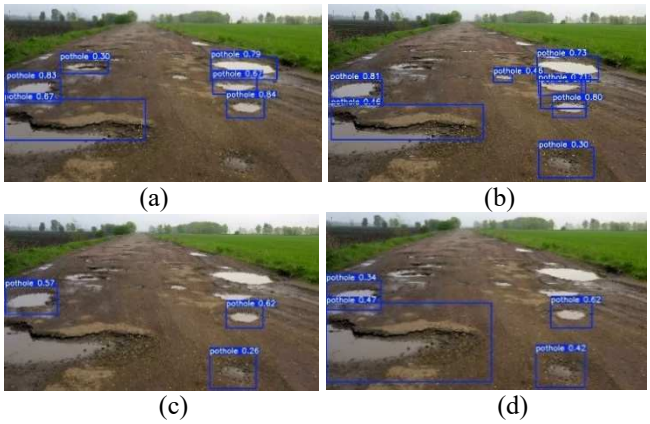


Fig. 8. (a)Default YOLOv5m Raw Image Detection, (b)YOLOv5m Raw Image Detection with Transpose Layer, (c)YOLOv8 Raw Image Detection, (d)YOLOv8 Raw Image Detection with Transpose Layer

In real-time scenarios, YOLOv5m (Fig. 8a) detected six potholes per frame with low latency (inference: 34.9 ms). When equipped with ConvTranspose2d (Fig. 8b), the detection increased to 8 potholes while maintaining inference speed (35.8 ms), reinforcing that transpose layers improve detection without compromising real-time capability in YOLOv5m. Conversely, YOLOv8m (Fig. 8c) achieved faster inference (51.4 ms) but detected only three potholes. With transpose layers (Fig. 8d), detection rose to 4, but inference slowed significantly to 70.2 ms. While the gain in recall is evident, the cost in speed must be considered, especially for deployment in real-time systems.

C. *Metrics Evaluation*

TABLE III. PERFORMANCE COMPARISON OF YOLO MODELS

Model	YOLOv5 [9]	YOLOv5 Transpose	YOLOv8 [13]	YOLOv8 Transpose
Parameter	20,852, 934	21,590, 790	25,840, 339	36,077, 539
Precision	0.847	0.894	0.888	0.840
Recall	0.760	0.720	0. 640	0.720
mAP50	0.826	0.839	0. 751	0.777
mAP50-95	0.420	0.417	0. 360	0.374

Table 2 summarizes the performance of four YOLO model variants: YOLOv5m, YOLOv5m with transpose layers, YOLOv8m, and YOLOv8m with transpose layers. Evaluation was conducted using Precision, Recall, mAP@50, and mAP@50-95. YOLOv5m with transpose layers showed consistent improvement, with Precision increasing from 0.847 to 0.894 and mAP@50 from 0.826 to 0.839, while Recall slightly decreased from 0.760 to 0.720. This reflects better localization performance with minimal reduction in detection coverage. YOLOv8m with transpose layers showed a different pattern, improving Recall from 0.640 to 0.720 and mAP@50-95 from 0.360 to 0.374, but with a drop in Precision from 0.888 to 0.840. This suggests that transpose layers help detect more varied or subtle objects but may introduce more false positives.

D. *Discussion and Analysis*

The findings confirm the hypothesis that transpose layers can enhance detection performance, especially for small and irregular objects. However, their impact differs according to the model architecture: YOLOv5m benefits from improved spatial representation without much latency penalty. YOLOv8m, while more modern, is more sensitive to changes in architecture, and transpose layers introduce overhead that may negate real-time advantages. This underscores the importance of aligning architectural enhancements with each model's design goals.

V. CONCLUSION AND FUTURE SCOPE

This study compared YOLOv5m and YOLOv8m, with and without transpose convolution layers, for pothole detection in static images and real-time videos. As a learnable upsampling method, adding transpose layers enhanced spatial feature reconstruction and improved the models' ability to detect small and irregular potholes under complex visual conditions.

YOLOv5m with transpose layers achieved a balanced performance, increasing mAP@50 from 0.826 to 0.839 (1.57%) with minimal impact on speed. YOLOv8m showed a larger mAP@50 gain from 0.751 to 0.777 (3.46%) and better mAP@50-95, though at the cost of increased latency and reduced precision. These results indicate that the effectiveness of transpose layers depends on the model architecture.

Detecting small and subtle defects like potholes is crucial for road safety and smart infrastructure. The learnable nature of transpose convolution enhances spatial detail, making it valuable not only for pothole detection but also for tasks such as crack detection, medical imaging, industrial inspection, and satellite analysis. Future research should focus on optimizing transpose layer settings to balance accuracy and efficiency in real-world applications.

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